

HEMA SAGAR KOPPUSETTI

AI/ML Engineer | Computer Vision | NLP | Generative AI
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PROFESSIONAL SUMMARY

Final-year B.Tech student specializing in AI/ML with hands-on research experience at IIT Tirupati and industry-grade certifications from NVIDIA, AWS, and Google. Demonstrated expertise across deep learning, computer vision, NLP, and ML-driven positioning systems. Ranked All India 4th in NVIDIA GPU Computing. Experienced in building end-to-end AI pipelines for real-world applications — from GNSS error correction to automated content intelligence. Eager to apply technical expertise and cross-cultural adaptability to support Japanese SME innovation and globalization goals.

EDUCATION

Lendi Institute of Engineering and Technology

B.Tech in CSE (Artificial Intelligence & Machine Learning) | CGPA: 8.79 / 10.0

Sep 2023 – May 2027

Vizianagaram, AP

Smt. Godavari Devi Saraf Senior Secondary School

Class XII, CBSE | Percentage: 85.0%

Jun 2021 – Mar 2023

Garividi, Andhra Pradesh

Smt. Godavari Devi Saraf Senior Secondary School

Class X, CBSE | Percentage: 90.0%

Apr 2020 – Mar 2021

Garividi, Andhra Pradesh

TECHNICAL SKILLS

Programming: Python, C, Java, SQL

ML Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM

Deep Learning: CNN, RNN, LSTM, TCN, Transformer, BERT, GPT, Vision Transformers, SimCLR

Generative AI & LLMs: Hugging Face, Prompt Engineering, Claude API, LLM Fine-tuning

Cloud & GPU: AWS (AI-Powered), Google Cloud (Vertex AI), CUDA, GPU Profiling

Data & Tools: NumPy, Pandas, OpenCV, Streamlit, Plotly, Matplotlib, Joblib, Git, Jupyter

AI Automation: n8n Workflow Automation, AI Agent Design, REST API Integration

Specializations: Computer Vision, NLP, Reinforcement Learning, GNSS/Positioning ML, Generative AI

RESEARCH EXPERIENCE

Research Intern — IITTNI STAR PNT, IIT Tirupati

Selected among 3,000+ applicants | NavIC/GNSS ML Error Correction System

Aug 2025 – Nov 2025

Tirupati, AP

[Project Link](#)

- Built an end-to-end ML pipeline correcting NavIC/GNSS positioning errors in Indian urban environments caused by NLOS signal loss, multipath propagation, poor satellite geometry, and orientation-dependent noise.
- Engineered signal-quality features from raw Android GNSS logs: Mean SNR, satellite count, elevation angle, azimuth, roll/pitch/yaw, inclination angle, and position jump.
- Trained and benchmarked 5 models — Random Forest, XGBoost, SVM, LSTM, TCN — evaluated on RMSE, MAE, R^2 , accuracy, precision, and recall across X and Y error dimensions.
- Random Forest achieved best RMSE (X: 1.243, Y: 1.794) and R^2 (X: 0.898, Y: 0.909), confirming tree-based models outperform sequence models on structured GNSS tabular data.
- XGBoost delivered the highest accuracy (99.27%) with competitive RMSE; SVM served as a strong non-linear classical baseline.
- LSTM and TCN showed negative R^2 values, indicating the dataset lacks the strong temporal continuity that sequence models require — a key insight for future GNSS ML research.
- Built an interactive Streamlit dashboard with dynamic model selection, raw vs. corrected trajectory visualization, and per-model error distribution plots for real-world GNSS correction deployment.

KEY PROJECTS

AI-Powered Business Intelligence & Automation for SMEs ★ METI-Relevant

Jan 2026 – Mar 2026

Python, n8n, Claude API, Google Sheets, YouTube Data API, Instagram
Graph API

[Project link](#)

- Designed and deployed a 25-node n8n workflow for viral content detection, AI-powered scoring, and cross-platform publishing — directly applicable to SME digital marketing automation.
- Integrated Claude AI (Haiku + Sonnet) for intelligent content analysis, script generation, and metadata optimization, reducing estimated manual effort by 80%.
- Built automated pipelines across YouTube, Instagram, Google Drive, and Gmail; delivered end-to-end ROI-focused automation with full error handling and API quota management.

Cross-Domain AI Reliability for Industrial Inspection ★ METI-Relevant

May 2025 – Jun 2025

Python, PyTorch, BERT, Hugging Face, CUDA —
github.com/HemaSagarKoppusetti

[Project link](#)

- Fine-tuned BERT-base-uncased for multi-domain sentiment classification; achieved 88.26% accuracy / 88.43% F1 on IMDB and quantified cross-domain degradation on SST-2 (85.89% accuracy / 87.01% F1).
- Applied mixed precision training, dynamic batch scaling, and pipeline parallelism to reduce GPU training time by 40% without accuracy loss.
- Delivered calibration analysis, reliability diagrams, and confusion matrix reports — directly usable for quality assurance systems in manufacturing environments.

Self-Supervised Visual Intelligence for Low-Label Environments

Mar 2025 – Apr 2025

PyTorch, Vision Transformers, SimCLR, ResNet-18, Contrastive
Learning

[Project Link](#)

- Implemented SimCLR contrastive self-supervised learning pipeline on CIFAR-10 using ResNet-18 encoder; trained for 50 epochs with NT-Xent loss (Adam, lr 3×10^{-4} , batch 128).
- Reduced training loss from 5.056 to 4.390; achieved 59.93% peak linear evaluation accuracy; evaluated robustness under 10%, 50%, 100% label conditions via t-SNE — validating label-efficient industrial inspection.

CERTIFICATIONS

- Google Cloud Data Analytics Certificate — Google Cloud (June 2026)
- AWS AI-Powered Cloud Engineer — AWS Educate (March 2026)
- AWS Cloud Certificate — AWS Academy (December 2025)
- Google AI ML Certificate — Google for Developers (September 2025)
- Google Cloud Generative AI — Google Cloud (June 2025)
- CUDA C/C++ and CUDA Python — NVIDIA (October 2024)

ACHIEVEMENTS & RECOGNITION

- All India Rank 4 — GPU Accelerated Computing & Codeathon (NVIDIA, 10-day national competition)
- Selected Research Intern — IIT Tirupati STAR PNT Challenge (Top among 3,000+ applicants nationwide)
- Special Prize — 24-Hour Vibe Coding Challenge, Cursors 2025, ANITS College
- Rank 48 Engineering / Rank 562 Globally — Unstop Platform (100,000+ participants)
- All India Rank 22,000 — GATE 2026 DA Paper, IIT Guwahati (among 69,000 participants)

LEADERSHIP & EXTRACURRICULAR

- Technical Lead — ACM Lendi Student Chapter (2024): organized AI/ML workshops, mentored 50+ students.
- Event Coordinator — LNIT Summit 2025 (National event, 1,500+ participants): managed technical sessions and logistics.
- Participant — Google DevFest 2024, GDG Vizag, GITAM University: exposure to global developer ecosystem.
- Attendee — 3-Day Cybersecurity Workshop by Cyber Rakshak Team: cross-domain technical exposure.

LANGUAGES

English (Professional Proficiency) **Telugu** (Native) **Japanese** (Beginner — actively learning for professional collaboration)